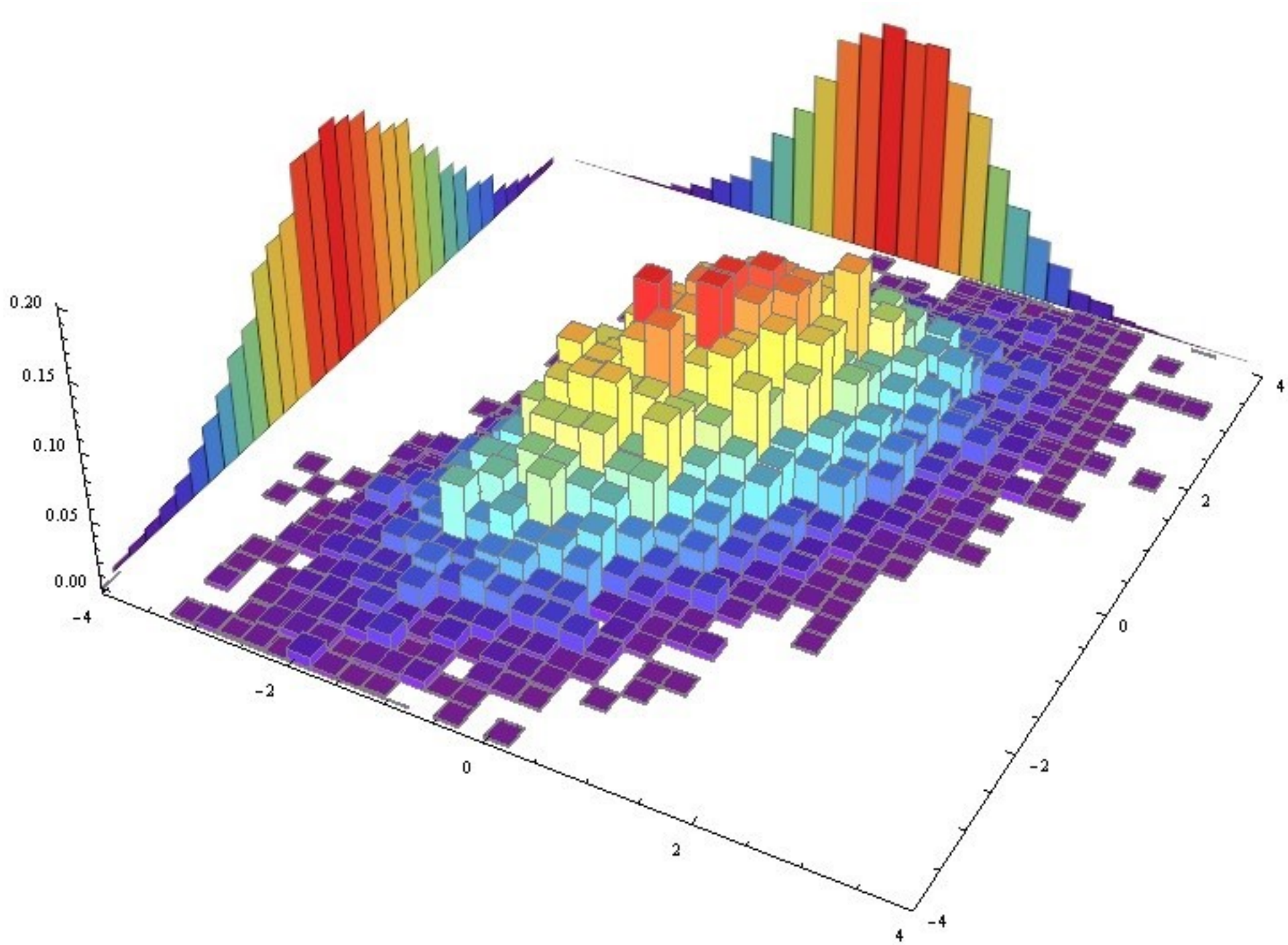
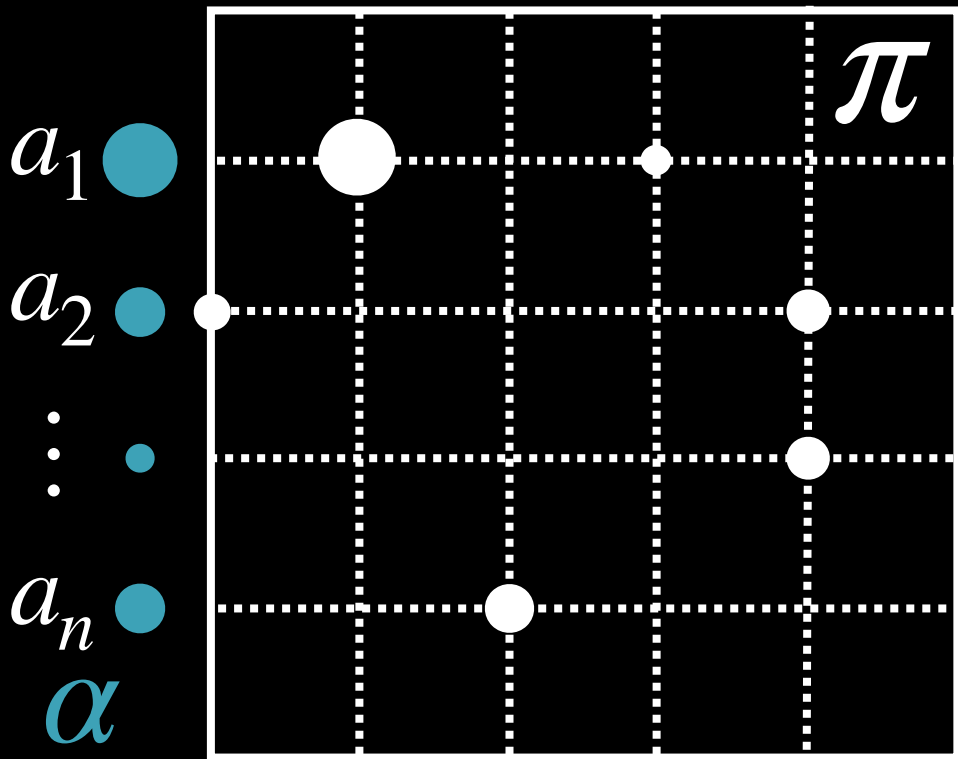




b_1 b_2 $\dots$ b_m β

$$\alpha = \sum_1^n a_i \delta_{x_i}, \quad a \in \Delta_n,$$

$$\beta = \sum_1^m \mathbf{b}_j \delta_{y_j}, \mathbf{b} \in \Delta_m$$



Couplings Formalize “Splitting”

Prob. measures:

$$\alpha = \sum_1^n \mathbf{a}_i \delta_{x_i}, \mathbf{a} \in \Delta_n,$$

$$\beta = \sum_1^m \mathbf{b}_j \delta_{y_j}, \mathbf{b} \in \Delta_m$$

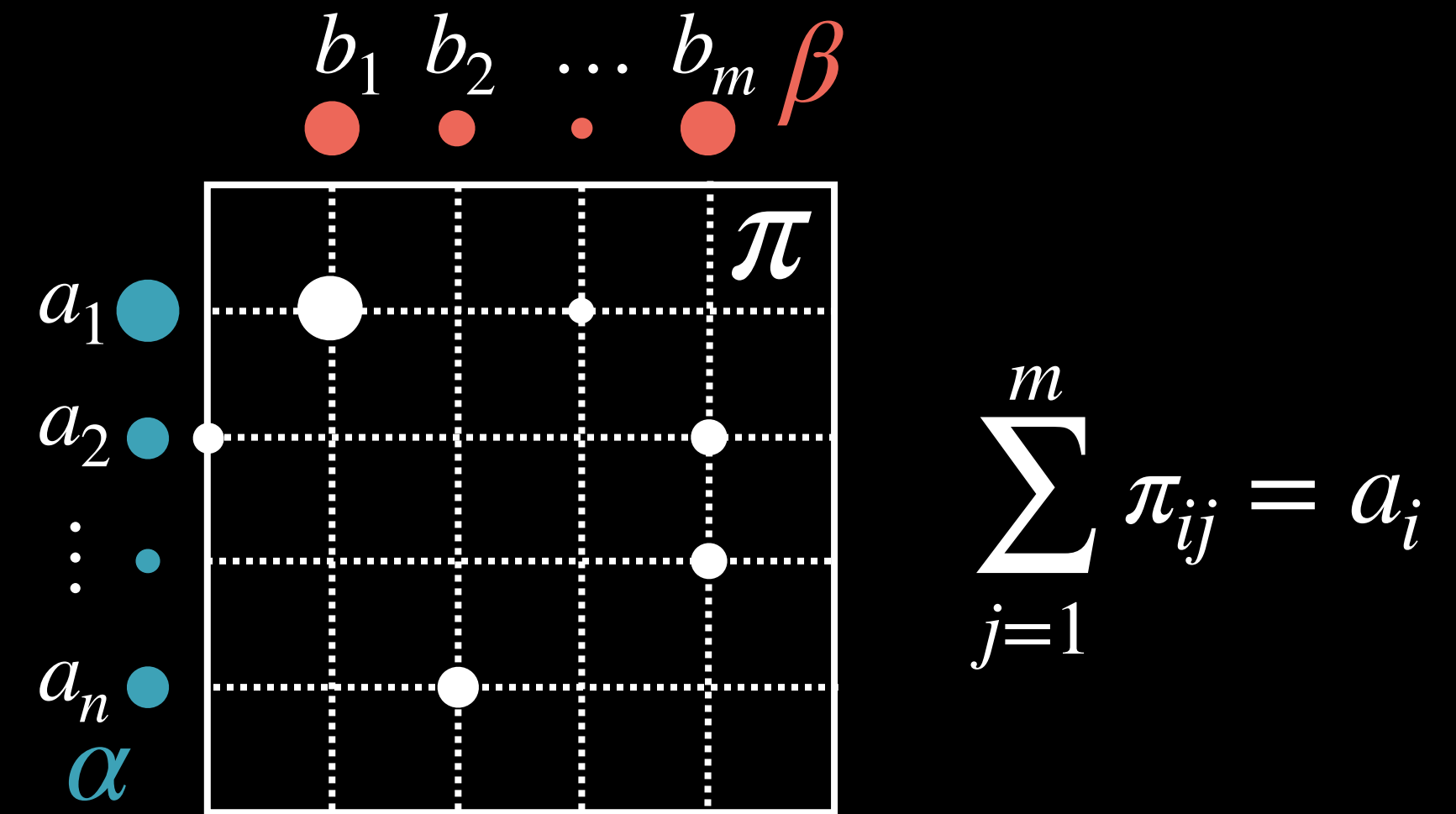
Coupling: $\pi \in \mathbb{R}_+^{n \times m}$

$\pi_{i,j}$ = mass transported from i to j

π is a **valid coupling** if it has marginals α, β
(i.e., its rows sum to $\mathbf{a}$ and its columns sum to $\mathbf{b}$)

The set of all admissible couplings:

$$\Pi(\mathbf{a}, \mathbf{b}) = \{ \pi \in \mathbb{R}_+^{n \times m} \mid \pi \mathbf{1}_m = \mathbf{a}, \pi^\top \mathbf{1}_n = \mathbf{b} \}$$



$$\sum_{i=1}^n \pi_{ij} = b_j$$



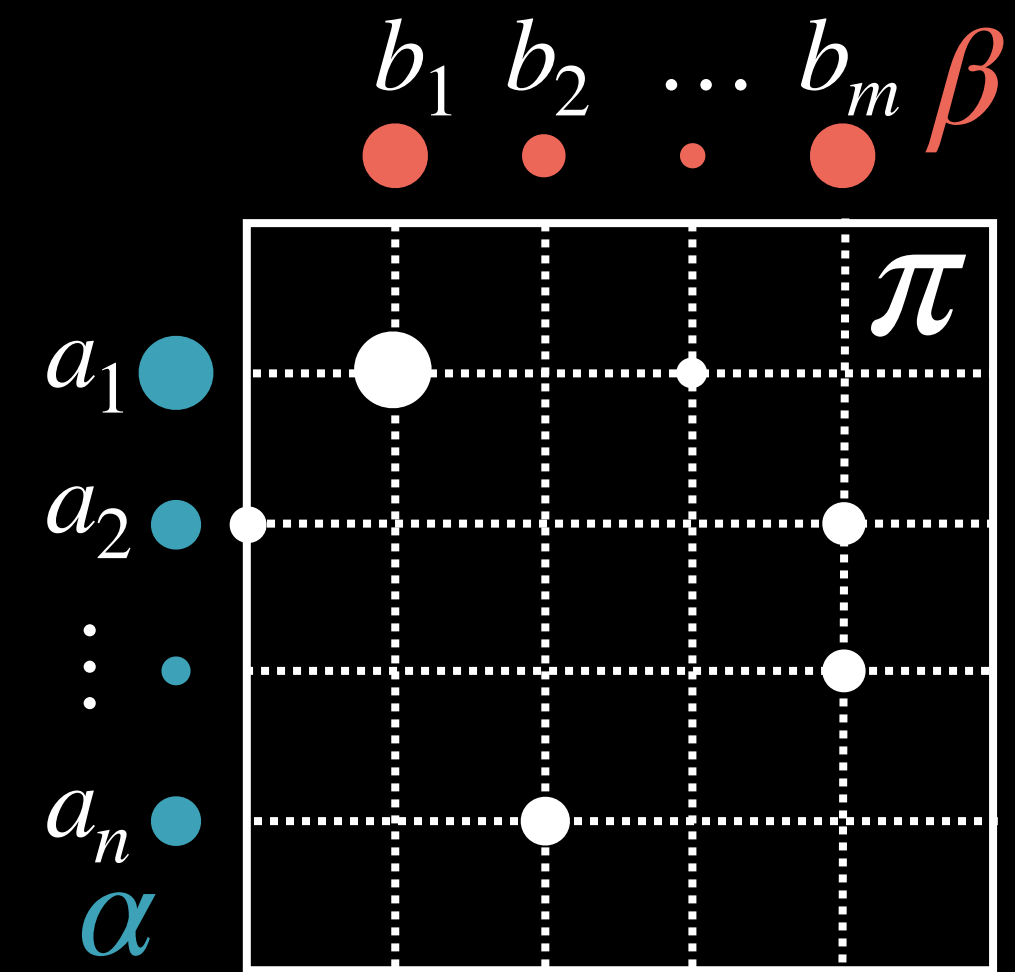
Transportation
Polytope

Pop-Quiz: Double Sums

Prob. measures:

$$\alpha = \sum_1^n \mathbf{a}_i \delta_{x_i}, \mathbf{a} \in \Delta_n,$$

$$\beta = \sum_1^m \mathbf{b}_j \delta_{y_j}, \mathbf{b} \in \Delta_m$$



$$\sum_{j=1}^m \pi_{ij} = a_i$$

$$\sum_{i=1}^n \pi_{ij} = b_j$$